

Cross-Family Predominance in the Homology Cliff: Preregistered Pfam Robustness Across Reference Panels and Mapper Topology

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April 12, 2026 (v1.2; confirmatory 10-seed extension 2026-08-24; figure erratum 2026-06-16)

Abstract

We partition distant-stratum false alarms of frozen ESM-2 t30 cosine k -NN retrieval by Pfam family membership. At the original $R=1000$, $k=25$ panel seed 20260410, 41 distant-stratum proteins receive a positive prediction while the true label is negative. Twenty are evaluable under the prespecified annotation rule, and all 20 are CROSS_FAMILY (zero shared Pfam identifiers with the positive-voting neighbors; per-seed Wilson 95% interval for the within-family fraction $[0, 0.161]$). A subsequently SHA256-locked confirmatory extension repeated the unchanged analysis across the ten already-fixed panel seeds 20260410–20260419. Cross-family false alarms predominated in every seed. The per-seed cross-family fraction had median 1.000, mean 0.994, and range 0.962–1.000; all ten seeds had nonzero evaluable denominators, satisfying the preregistered strong robustness criterion. Across seeds, 102 unique accessions were evaluable: 101 were always cross-family, one was always within-family, and none switched status. The seed, not the recurring accession appearance, is the preregistered robustness unit. Complementary Mapper decomposition provides exploratory geometric context but is not a causal test. The combined evidence disfavors a purely within-family panel-coverage explanation for the cliff in this tested regime; it does not imply that within-family errors are absent or that Pfam exhausts biological relatedness. For deployment, within-family panel expansion should not be relied on as the sole rescue; distant positive predictions still require review, and the panel-only learned projection from Paper 1 is the supported intervention tested here.

Erratum (2026-06-16). The Mapper scatter (Fig. 1) is now generated directly from the committed Mapper graph (`data/results_summaries/mapper_graph.json`): it plots the dominant node per occupied PCA lens bin, colored by positive fraction. The previously plotted point values were hand-entered and did not all match the committed graph; the Pfam partition results and all numerical claims are unchanged.

1 Setup

Test set: 24,885 proteins, 7,133 positive (28.7%). Embeddings: frozen ESM-2 t30 (150M parameters, 640d), L_2 -normalized. Every panel contains 1,000 class-balanced proteins; $k = 25$; distant stratum $s_{\max} < 0.90$; positive prediction when $\text{votes} \cdot 2 \geq k$. Pfam annotations cover 21,615 of 24,885 accessions (86.9%) in `data/annotations/proteins_25k_pfam.json`. The confirmatory extension changes only the reference-panel seed. All ten seeds, the evaluability rule, Pfam partition, aggregation, and decision rule were locked in `data/prereg/PRE_REGISTRATION_CROSS_FAMILY_10SEED_v1.md` before any additional seed was run.

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2 Single-seed result: zero within-family false alarms

quantity	count
distant stratum ($s_{\max} < 0.90$)	154
distant false alarms (pred=1, true=0)	41
false alarms with query Pfam known	36
false alarms with ≥ 1 positive-voter Pfam known	20
within-family (shared Pfam with voters)	0
cross-family (zero shared Pfam)	20

Every one of the 20 evaluable distant false alarms at seed 20260410 has zero Pfam overlap with the panel members that voted it positive. This original result remains unchanged and is committed in `data/results_summaries/cross_family_partition.json`.

3 Confirmatory robustness across preregistered panel seeds

seed	distant	distant FP	evaluable	within	cross	cross fraction
20260410	154	41	20	0	20	1.000
20260411	187	67	30	0	30	1.000
20260412	174	100	49	1	48	0.980
20260413	187	52	27	0	27	1.000
20260414	169	56	26	1	25	0.962
20260415	175	50	24	0	24	1.000
20260416	165	64	44	0	44	1.000
20260417	163	49	23	0	23	1.000
20260418	162	63	31	0	31	1.000
20260419	175	57	30	0	30	1.000

Across ten preregistered reference-panel seeds, cross-family errors predominated in every seed. The per-seed cross-family fraction had a median of 1.00, mean of 0.994, and range 0.962–1.00, satisfying the preregistered criterion for robustness to reference-panel composition. No seed had a zero evaluable denominator.

The secondary accession summary contains 102 unique evaluable accessions: 101 are always CROSS_FAMILY, one (Q5DFZ7) is always WITHIN_FAMILY in the seeds where it is evaluable, and none are mixed. The two within-family appearances of Q5DFZ7 occur in seeds 20260412 and 20260414 and share PF05980 in both. Across all seeds there are 304 evaluable appearances (302 cross-family, 2 within-family), but those appearances are not independent because accessions recur. Consistent with the preregistration, we therefore do not pool them into a binomial confidence interval; the panel seed is the robustness unit. The full per-seed and per-accession audit trail is committed in `data/results_summaries/cross_family_partition_10seed.json`.

4 What this disfavors

A natural explanation for the cliff is a within-family coverage gap: a distant query from family F lacks a close panel homolog from F , so adding more members of F should repair retrieval. The original 20/20 result and the preregistered ten-seed panel-composition extension both disfavor that explanation as the dominant mechanism in this dataset and fixed $t30/R=1000/k=25$ regime.

Cross-family false alarms outnumber within-family false alarms in every panel seed, and the worst observed seed is still 25/26 (0.962) cross-family among evaluable cases.

The result is not categorical. Within-family errors do occur, as shown by Q5DFZ7 in two seeds. Pfam-ID intersection is also a coarse family definition and cannot exclude shared superfamily, fold, or other biological relationships. The supported claim is therefore cross-family predominance robust to the tested panel-composition perturbation, not that every cliff error is cross-family or that within-family panel expansion can never help an individual case.

5 Mapper context

Mapper decomposition on PCA-2 lens following Singh, Mémoli, and Carlsson [1–3] (10×10 cover, 30% overlap, DBSCAN $\varepsilon=0.05$): 149 nodes, 6 entirely positive, 30 positive-enriched ($\text{pos_frac} > 0.5$). Positive class is contiguous on the manifold: top two positive-enriched nodes at PCA bins (7, 4) and (7, 3) contain 1,708 and 1,568 proteins at 88.5% and 93.9% positive rate.

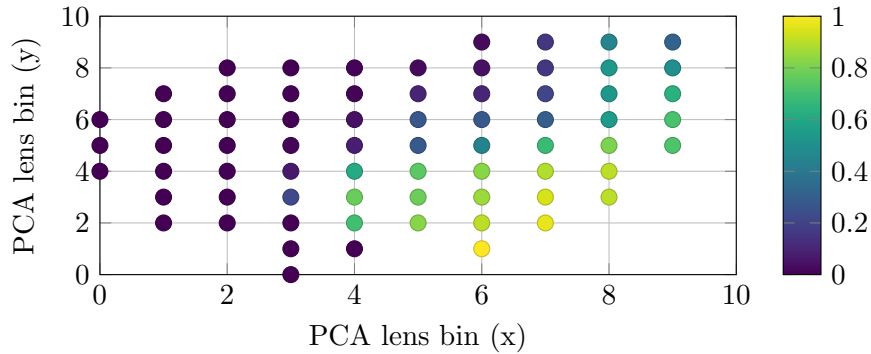


Figure 1: **Mapper 2D positive-fraction scatter.** Dominant Mapper node per occupied PCA lens bin (largest node by membership), colored by positive fraction pos_frac . The positive class localizes to a contiguous high-PC1 region ($\text{lens-}x \gtrsim 4$), peaking near bins (6–8, 1–4) at > 0.88 positive fraction; low-PC1 bins ($\text{lens-}x \leq 2$) are near-zero.

This spatial localization is consistent with, but does not prove, an embedding-space explanation for the Pfam result: positive proteins from multiple families occupy a concentrated manifold region, and distant false-positive queries can retrieve residents of that region despite having no shared Pfam identifier. Mapper is used here as exploratory context only; the confirmatory claim rests on the preregistered Pfam partition across panel seeds, not on a causal interpretation of the Mapper graph.

6 Biosecurity deployment consequence

The ten-seed confirmation strengthens the case against relying on within-family panel expansion as a generic cliff rescue in this tested regime. It does not rule out benefit for a particular query or a different family ontology. A deployer should therefore treat panel expansion as insufficient evidence of safety rather than as a guaranteed failure.

Two actionable consequences follow:

1. Route every distant-stratum positive prediction to human review, regardless of vote count or s_{max} . Calibration collapse (Paper 3) independently supports this rule.

2. Apply the panel-only learned linear projection (Paper 1) as preprocessing before retrieval. At t_{30} $R=1000$ $k=25$, the projection improves distant F_1 from 0.120 to 0.177 (47% relative). This is an empirically tested intervention on the embedding geometry, whereas within-family panel expansion is not supported as a general rescue by the Pfam robustness result.

7 Limitations

(L1) The confirmatory extension tests robustness only to reference-panel composition at fixed ESM-2 t_{30} , $R = 1000$, $k = 25$, threshold 0.90, dataset, labels, and Pfam ontology; it does not establish invariance across models, k , thresholds, or datasets. (L2) 13% of test proteins lack Pfam annotations and are excluded from the WITHIN/CROSS denominator under the locked evaluability rule. (L3) Pfam-ID overlap is a coarse family definition. InterPro superfamily, CATH fold, or structure-aware relations could identify biological relatedness that the Pfam test misses. (L4) The result at $k = 25$ need not hold at $k = 1$ or under a different voting rule. (L5) This is a single-dataset, single-author result and has not yet been independently replicated by a third party.

8 Data availability

The original single-seed per-case partition is `data/results_summaries/cross_family_partition.json`. The preregistered ten-seed result, including every per-seed count, full distant-false-positive detail, seed-level aggregate, and per-accession history, is `data/results_summaries/cross_family_partition_10seed.js` (SHA256 332b939622ac2a17...). The locked protocol is `data/prereg/PRE_REGISTRATION_CROSS_FAMILY_10SEED` (SHA256 204953efeb098f70...).

References

- [1] Gurjeet Singh, Facundo Mémoli, and Gunnar Carlsson. Topological methods for the analysis of high dimensional data sets and 3D object recognition. *Eurographics Symposium on Point-Based Graphics*, 2007.
- [2] Gunnar Carlsson. Topology and data. *Bulletin of the AMS*, 46(2):255–308, 2009.
- [3] Herbert Edelsbrunner and John L Harer. *Computational topology: an introduction*. AMS, 2010.